

Alpha and Beta Diversity in Microbiome Analysis

Core ecological concepts for
metagenomics and microbiome studies

MMB Metagenomics

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Learning Objectives

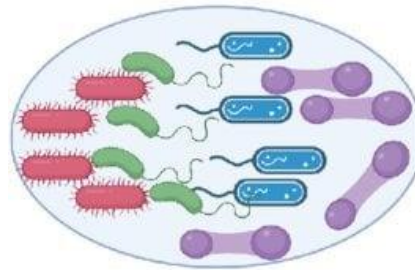
- Define alpha and beta diversity
- Understand common diversity metrics
- Interpret diversity plots and analyses
- Recognize strengths and limitations of different methods

Why Diversity Matters

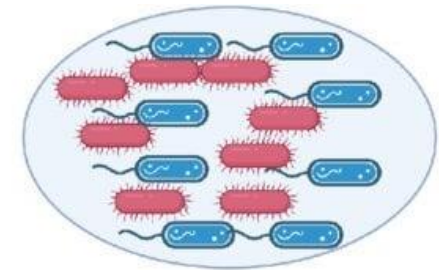
- Microbial communities vary across environments
- Diversity reflects ecosystem structure and stability
- Used to compare healthy vs diseased states
- Important in ecology, medicine, and environmental science

What is Alpha Diversity?

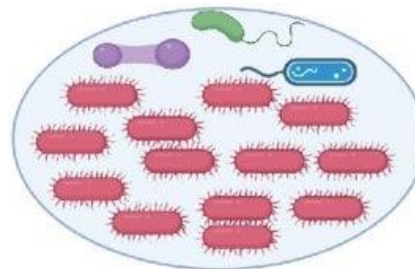
- Measures diversity within a single sample
- Describes richness and evenness
- Answers: 'How diverse is this sample?'



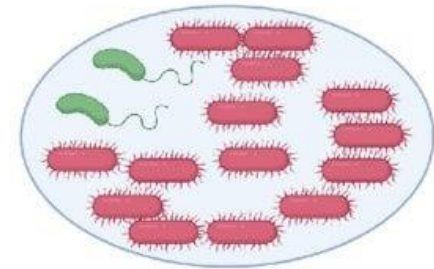
A. High richness and diversity (alpha diversity - high)



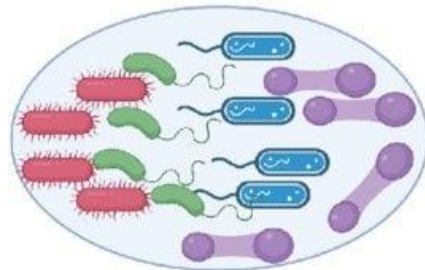
B. Low richness and diversity (alpha diversity - low to moderate depending on metric used)



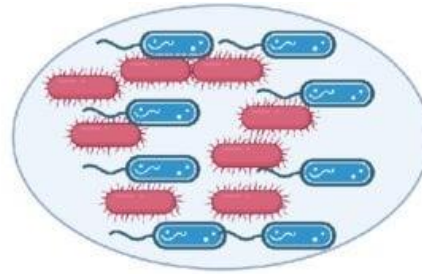
C. High richness, low diversity (alpha diversity - depends on metric used*)



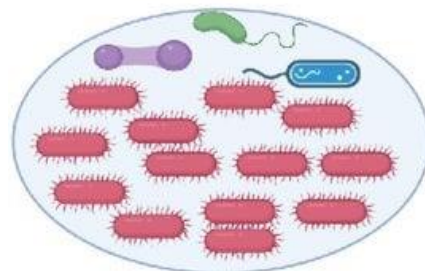
D. Low richness and diversity (alpha diversity - low)



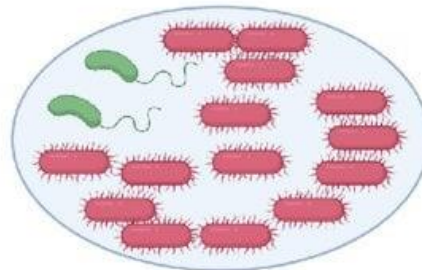
A. High richness and diversity (alpha diversity - high)



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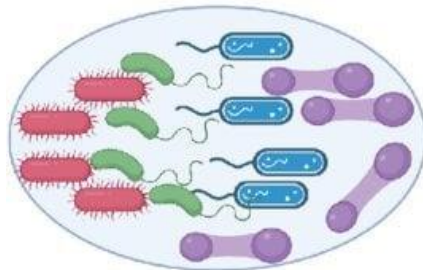
C. High richness, low diversity (alpha diversity - depends on metric used*)



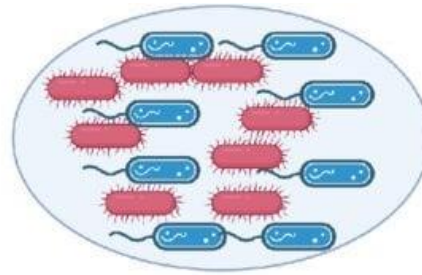
D. Low richness and diversity (alpha diversity - low)

Richness

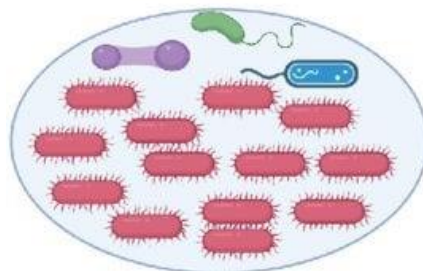
- Richness = number of observed taxa
- Higher richness means more unique species/OTUs/ASVs
- Does not consider abundance differences



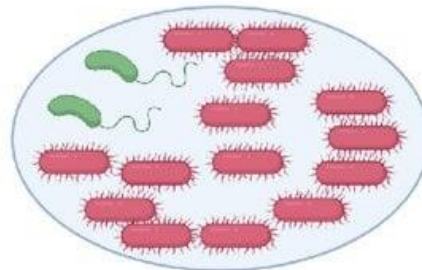
A. High richness and diversity (alpha diversity - high)



B. Low richness and diversity (alpha diversity - low to moderate depending on metric used)



C. High richness, low diversity (alpha diversity - depends on metric used*)



D. Low richness and diversity (alpha diversity - low)

Evenness

- Evenness describes how uniformly taxa are distributed
- High evenness = taxa have similar abundances
- Low evenness = few taxa dominate community

Observed Species Metric

- Counts number of observed taxa
- Simple richness estimate
- Sensitive to sequencing depth

Shannon Diversity Index

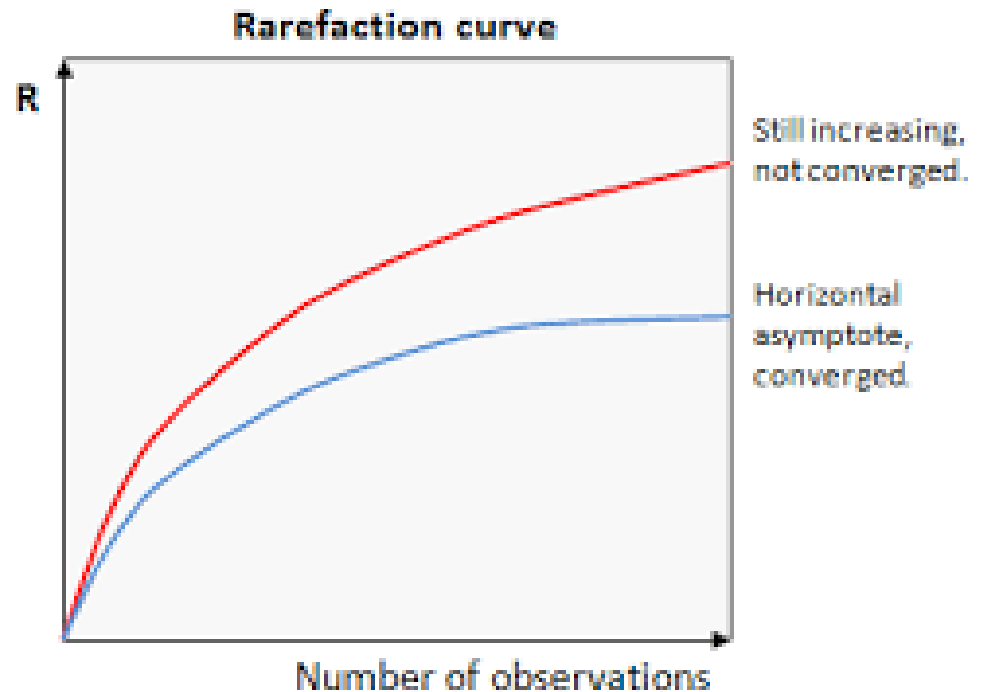
- Combines richness and evenness
- More sensitive to rare taxa
- Common in microbiome studies

Simpson Diversity Index / Inverse Simpson

- Simpson Diversity index measures dominance within a community
- More influenced by abundant taxa
- Lower sensitivity to rare organisms
- Usually $1 / D$ or the inverse Simpson index is used as the numeric value is more intuitive with the other diversity indices

Rarefaction Curves

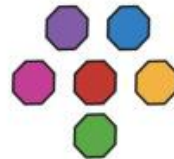
- Show species richness vs sequencing depth
- Used to evaluate sampling completeness
- Plateau suggests sufficient sequencing depth



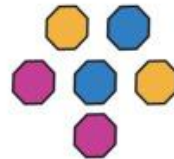
What is Beta Diversity?

- Measures differences between samples
- Compares community composition across groups
- Answers: 'How similar are these samples?'

Alpha Diversity:
Biodiversity with a microbial sample



High Alpha Diversity



Low Alpha Diversity

Beta Diversity:
Similarity between microbial samples



3 out of 6
microbial species
shared



Healthy Gut Microbiome vs
Dysbiotic Gut Microbiome
(PCOS)



Healthy Gut Microbial Sample



Dysbiotic Gut Microbial Sample
(PCOS)

Note: Each different colour represents a different microbial species

<https://doi.org/10.1016/B978-0-323-91190-0.00009-6>
Oshinjo et al., 2024

Types of Beta Diversity Metrics

- Presence/absence based metrics (often most robust results!)
- Abundance-based metrics
- Phylogenetic metrics

Jaccard Distance

- Based only on presence/absence
- Ignores abundance information
- Useful for simple community overlap

Bray-Curtis Dissimilarity

- Uses abundance information
- Very common in microbiome studies
- Sensitive to dominant taxa

UniFrac Distances

- Phylogeny-based beta diversity metric
- Weighted UniFrac uses abundance
- Unweighted UniFrac uses presence/absence
- Incorporates evolutionary relationships

Ordination Methods

- Reduce complex distance matrices into 2D plots
- Examples: PCoA, NMDS, PCA
- Samples clustering together are more similar

PCoA Plots

- Principal Coordinates Analysis
- Visualizes beta diversity distances
- Axes explain variation between samples

Interpreting Clustering

- Clusters may indicate biological similarity
- Separation may reflect treatment or environment
- Statistical tests needed for significance

PERMANOVA

- Statistical test for beta diversity differences
- Tests whether groups differ significantly
- Commonly implemented in vegan/adonis

Factors Affecting Diversity Estimates

- Sequencing depth
- DNA extraction bias
- Choice of classifier and database
- Normalization methods

Normalization Approaches

- Rarefaction
- Relative abundance scaling
- CLR transformation
- Variance stabilization methods

Alpha vs Beta Diversity

- Alpha diversity = within-sample diversity
- Beta diversity = between-sample diversity
- Both provide complementary ecological insights

Common Software Tools

- QIIME 2
- phyloseq (R)
- mia (R)
- vegan (R)
- scikit-bio (Python)

Practical Recommendations

- Use multiple diversity metrics
- Always inspect sequencing depth
- Interpret ordination carefully
- Combine ecological and statistical interpretation

Summary

- Alpha diversity measures diversity within samples
- Beta diversity compares samples
- Different metrics emphasize different biological patterns
- Correct interpretation requires biological context

Common Diversity Metrics

Metric	Type	Uses Abundance?	Main Feature
Observed Species	Alpha	No	Richness only
Shannon	Alpha	Yes	Richness + evenness
Inverse Simpson	Alpha	Yes	Dominance sensitive
Jaccard	Beta	No	Presence/absence
Bray-Curtis	Beta	Yes	Abundance differences